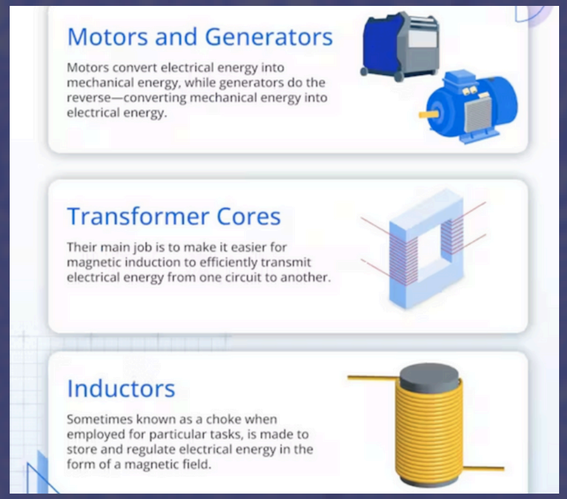
A Brief History

Silicon steel was first used for electrical applications in the **late 1800s**, when English metallurgist **Sir Robert Hadfield,** who discovered manganese, found that silicon increased the magnetic permeability of iron. The correlation between magnetic permeability and crystallographic orientations in iron single crystals was discovered in 1926 by Honda and Kaya, which led to subsequent developments to silicon steel. It wasn't until 2008 that Baosteel-after inventing the technologies and machinery for high-grade, effective non-oriented silicon steel production—managed to accomplish a RAPID, ALL-ENCOMPASSING PRODUCT UPGRADE.

Electrical steel *is speciality steel used in the cores of* ***electromagnetic*** *devices such as motors, generators, and transformers because it reduces power loss. It is an iron alloy with* ***silicon*** *as the main additive element (instead of carbon).*

- ## application



- ## *manufacturing*

electrical steel is classified into two main types: **Grain-Oriented (GO) and Non-Grain-Oriented (NGO)**. GO steel has oriented grains in a specific direction and NGO steel has randomly oriented grains.
Both types begin with melting iron/scrap with 1-6.5% silicon, casting into slabs, and hot rolling to 2-3mm. For NGO steel, the process continues with direct cold rolling (0.2-0.35mm), annealing (750-850°C), and coating, producing isotropic steel for motors/generators. GO steel requires additional steps: after hot rolling, it undergoes hot band annealing, followed by two-stage cold rolling with intermediate annealing, decarburization, and a critical high-temperature annealing (1200°C with MgO coating) to develop the Goss texture for transformer applications, finishing with a tension coating.t

6.1 Properties

(a) Composition. Added silicon can range from near zero up to 3+ %. In general higher silicon which restrains eddy currents reduces loss, is harder and costs more to make. Resistivity and as-annealed hardness varies with the percentage of silicon. Added silicon reduces B_{sat} and impairs high-field permeability. The absence of carbon, sulphur and titanium is desirable; desulphurisation is possible but expensive.

(b) Thickness. The thinner the steel the more effectively eddy currents are restrained and the lower the core losses. However a lamination pile of a given height contains less metal if made of 0.35 mm steel rather than of 0.65 mm steel due to the effect of the extra surfaces on the stacking factor. The effect of this is to reduce the effective B_{sat}. Thinner steel costs more to produce and requires more press-blows to make lamination stacks of a given size.

(c) Coated or not. Coatings are expensive to apply. Needless coating reduces the space factor and may cause weldability problems.

(d) State of heat treatment. Material can be finally annealed or semi-processed and awaiting a final anneal after stamping.

(e) Hardness. Material can be soft due to being finally annealed and having very little alloy content, e.g. VPN_{10} 90, or, progressively harder, as:

- Silicon is added up to 3%
- Phosphorous is added up to 0.08%
- Temper rolling is applied.

Hardness up to VPN_{10} 180 is acceptable. Note: the subscript 10 refers to a test done with a 10 kg load.

creators:Rasul alitabar malekshah-Rasol motamed-Aynaz chabouk

background : Non-oriented electrical silicon steel (image made with magneto-optical sensor and polarizer microscope)